

Project 2: Exploratory Data Analysis (EDA)
Dataset: E-commerce Sales Dataset
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Organization: DecodeLabs

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: plt.figure(figsize=(10,6))
sns.set_style("whitegrid")

<Figure size 1000x600 with 0 Axes>
```

```
[4]: df = pd.read_csv(r"C:\Users\zulkar\OneDrive\Desktop\Project_2_EDA\raw_dataset.csv")
df.head()
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	R
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	

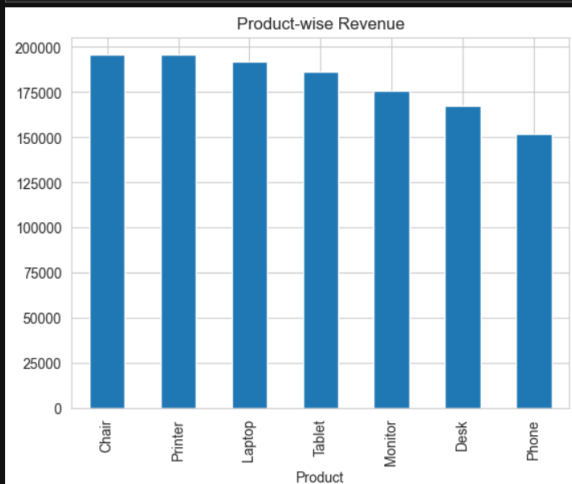
```
##Exploratory Data Analysis
```

```
[7]: df[['Quantity','UnitPrice','TotalPrice']].describe()
```

	Quantity	UnitPrice	TotalPrice
count	1200.000000	1200.000000	1200.000000
mean	2.945833	356.412750	1053.968300
std	1.407557	197.177146	819.856558
min	1.000000	11.390000	11.390000
25%	2.000000	186.062500	410.520000
50%	3.000000	364.210000	823.615000
75%	4.000000	521.570000	1578.475000
max	5.000000	699.930000	3456.400000

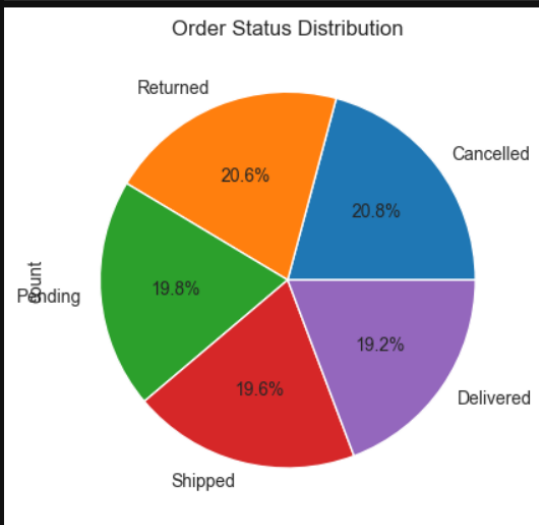
```
[8]: product_sales = df.groupby('Product')['TotalPrice'].sum().sort_values(ascending=False)
```

```
product_sales.plot(kind='bar')
plt.title("Product-wise Revenue")
plt.show()
```



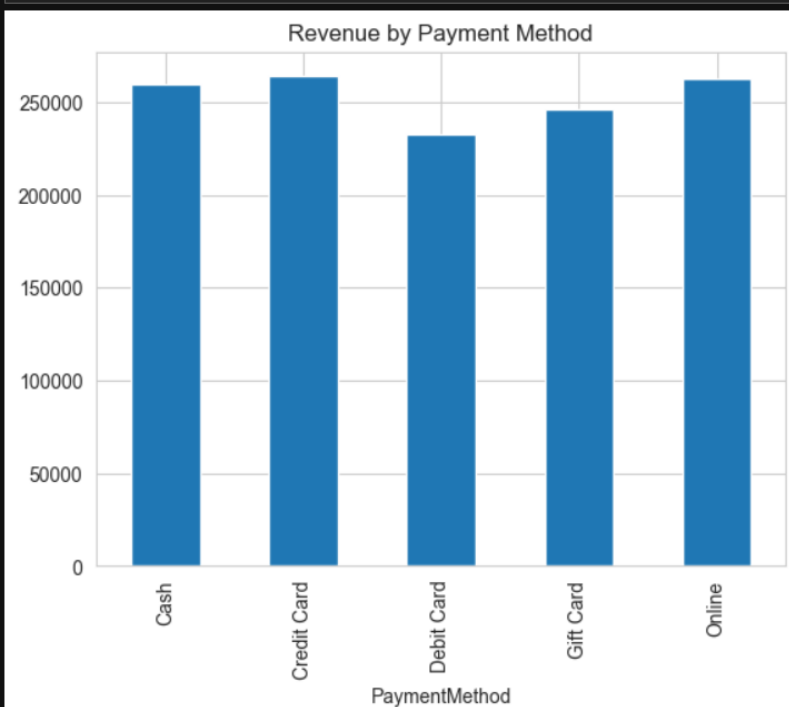
```
##Order Status Analysis
```

```
df['OrderStatus'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.title("Order Status Distribution")  
plt.show()
```



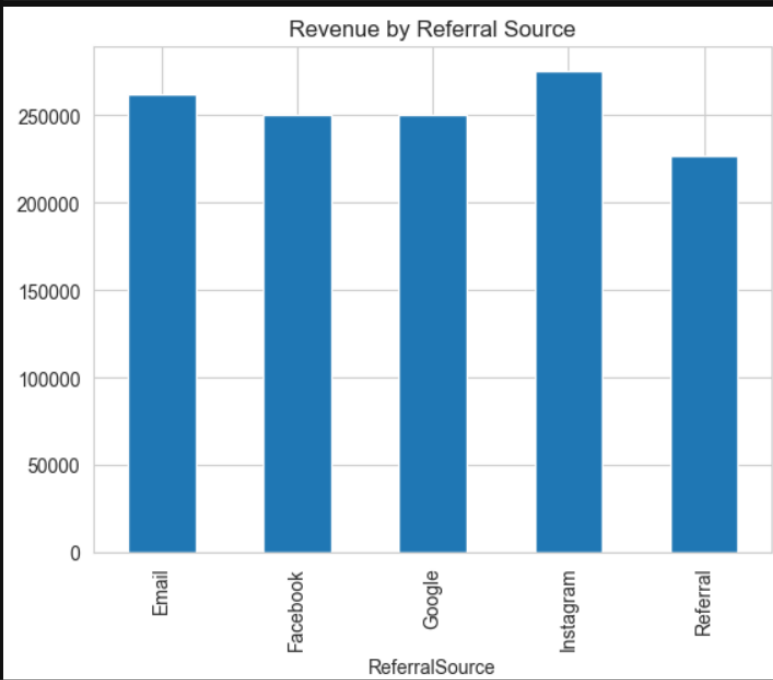
```
##Payment Method Analysis
```

```
df.groupby('PaymentMethod')['TotalPrice'].sum().plot(kind='bar')  
plt.title("Revenue by Payment Method")  
plt.show()
```



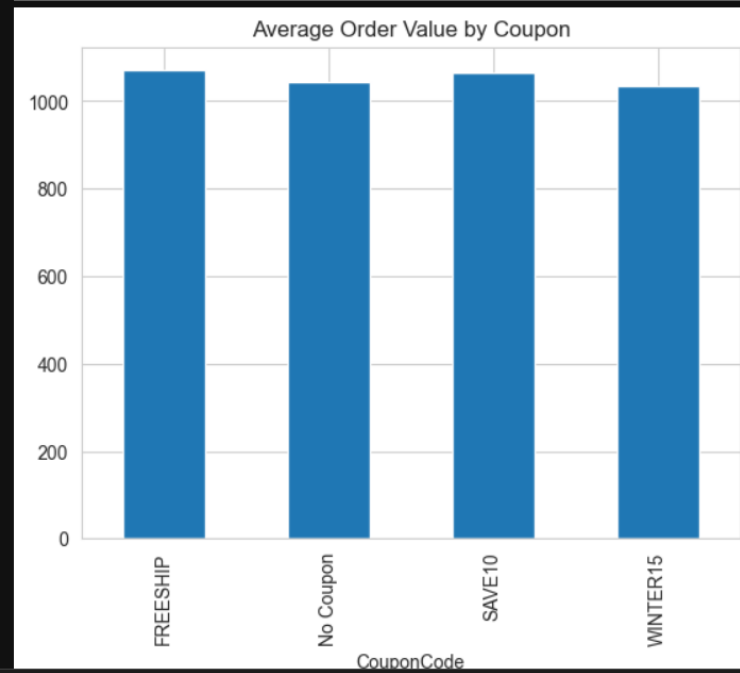
```
##Referral Source Analysis
```

```
1]: df.groupby('ReferralSource')['TotalPrice'].sum().plot(kind='bar')  
plt.title("Revenue by Referral Source")  
plt.show()
```



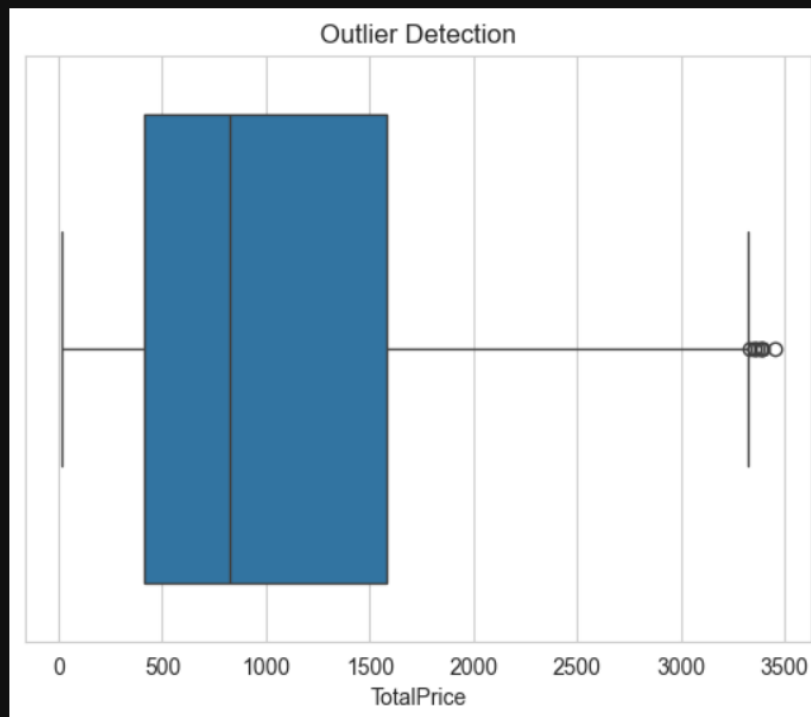
```
##Coupon Code Analysis
```

```
22]: df.groupby('CouponCode')['TotalPrice'].mean().plot(kind='bar')  
plt.title("Average Order Value by Coupon")  
plt.show()
```



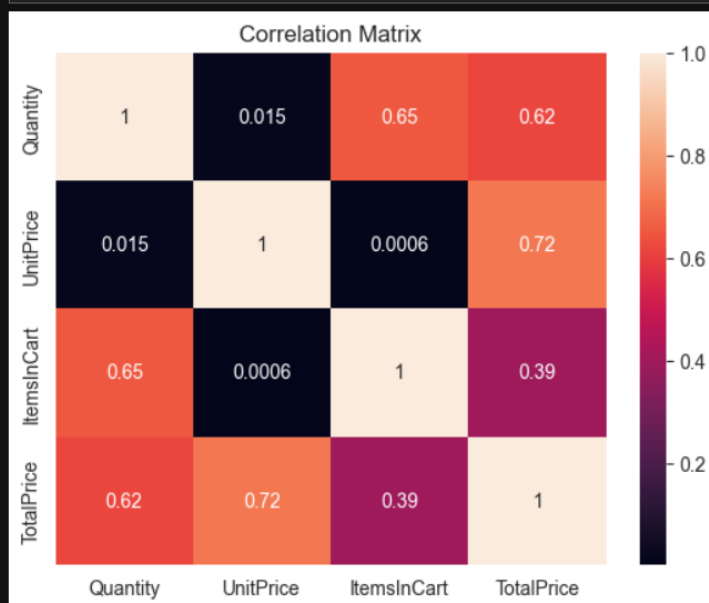
```
##Outlier Detection
```

```
3]: sns.boxplot(x=df['TotalPrice'])  
plt.title("Outlier Detection")  
plt.show()
```



```
##Correlation Heatmap
```

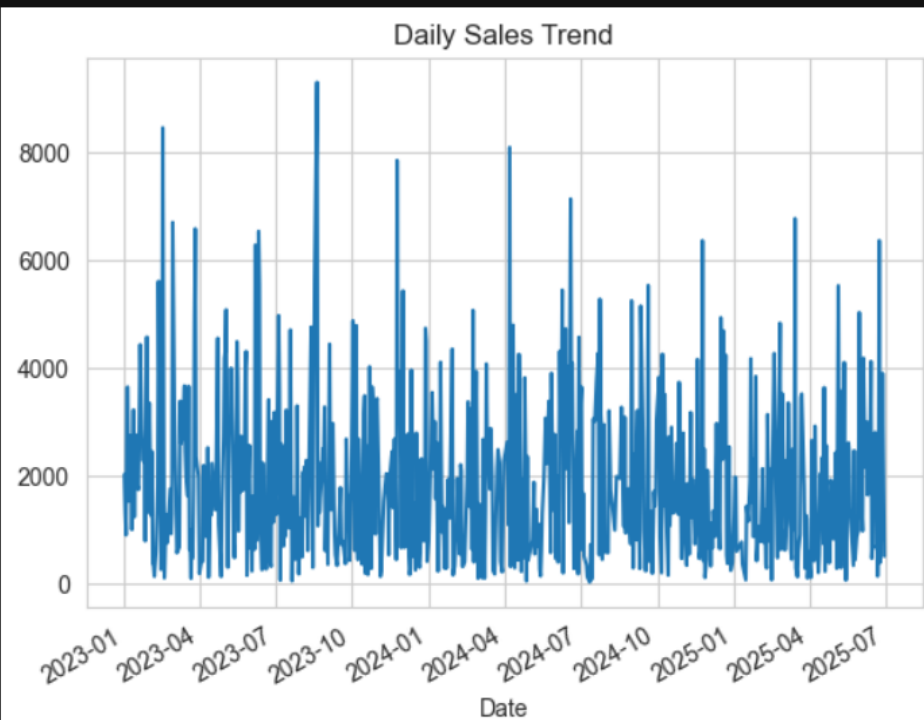
```
]: sns.heatmap(df.corr(numeric_only=True), annot=True)  
plt.title("Correlation Matrix")  
plt.show()
```



```
##Time Trend Analysis
```

```
daily_sales = df.groupby('Date')['TotalPrice'].sum()
```

```
daily_sales.plot()  
plt.title("Daily Sales Trend")  
plt.show()
```



#Key Insights

##1.chair generated highest revenue.

##2.Instagram is the strongest referral source.

##3.Credit card is the preferred payment method.

##4.Some extreme high-value orders exist.

##5.Sales show fluctuations over time.